

Network Forensics Week 12:

Linux Log Analysis

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Network Forensics

April 16, 2026

Abstract

This report documents a forensic investigation conducted on a Linux system using a set of five audit log files (audit.log through audit.log.4) and the journalctl system journal utility. The investigation was tasked with identifying unauthorized SSH access, tracing the user responsible for opening a suspicious file, uncovering evidence of malware staging in a hidden directory, investigating activity by a potentially compromised user account, and quantifying access to a specific log file. Using standard Linux command-line utilities including grep, cut, sort, uniq, and journalctl, the investigation identified two users who accessed the system via SSH, linked a suspicious HTML file to a specific user account, discovered malware staging activity in a hidden .backup directory containing nmap and shell script files, documented elevated privilege usage by the user 'student', and confirmed 364 accesses to the target status.log file. Findings are supported by highlighted terminal output and mapped to actionable investigative recommendations.

Introduction

Objective / Purpose

The objective of this assignment is to demonstrate proficiency in Linux log analysis as applied to a digital forensics scenario. Specifically, the investigation aims to extract actionable intelligence from Linux audit logs and the systemd journal by answering five targeted forensic questions relating to SSH access, suspicious file activity, malware staging, user behavior, and file access frequency.

Background Information

Linux systems generate two primary sources of activity logs relevant to this investigation. The Linux Audit Framework (auditd) captures kernel-level security events and writes them to /var/log/audit/audit.log, with older logs rotated to audit.log.1 through audit.log.4. These logs record events such as file access, process execution, network connections, and authentication using structured key-value fields including type, msg (timestamp), uid, auid, and name.

The systemd journal, accessed via journalctl, captures user-space service events including PAM authentication, session management, and sudo command execution. Unlike auditd, the journal does not use the audit log format and must be queried separately.

Together, these two sources provide complementary coverage of system activity. Audit logs are well suited for file and process tracing at the kernel level, while journalctl is better suited for tracking user session events, login history, and executed commands. Both are essential tools in a Linux forensic investigation.

Tools and Methods

Tools Used

- grep: Pattern-based text search across multiple files
- cut: Field extraction from delimited output
- sort: Alphabetical and numeric ordering of results
- uniq -c: Deduplication and frequency counting
- wc -l: Line count for occurrence totals
- journalctl: Systemd journal query utility (with sudo for full access)
- Linux Bash terminal (student@analystworkstation)

Procedure

All five audit log files were downloaded and placed in a single directory (~Downloads/audit_logs/) so that the glob pattern audit.log* could search across all files simultaneously. The -h flag was used with grep to suppress filenames in output, producing cleaner results.

For Tasks 1 through 3, grep commands were chained with pipe operators (|) to progressively filter and refine output. The -o flag extracted only the matching portion of each line, and -oP enabled Perl-compatible regex for named field extraction. Results were piped through sort and uniq -c to produce frequency-sorted summaries.

For Task 4, journalctl required sudo privileges to access the full journal. Output was first piped directly to grep for quick filtering, then exported to a file (/tmp/journal_export.txt) using redirection for repeated searching without re-querying the journal daemon.

For Task 5, the exported journal file was searched with grep and the result piped to wc -l to count total occurrences of the target filename. The audit logs were also checked for comparison.

Results

Finding 1: SSH Users Who Accessed the System

Information Given	A set of five audit log files covering system activity on the target Linux machine.
Task	Identify all users who successfully authenticated via SSH.

Result	Two users: cbarton (56 sessions) and bmorse (34 sessions).
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The following command was run to search all audit log files for successful SSH authentication events and summarize unique usernames with their occurrence counts:

```
grep -h "ssh" audit.log* | grep "res=success" | grep -o 'acct="[^\"]*"' | sort |
uniq -c | sort -rn
```

```
student@analystworkstation:~/Downloads/audit_logs
student@analystworkstation Downloads$ cd audit_logs/
student@analystworkstation audit_logs$ ls -lh audit.log
-rw-r--r-- 1 student student 4.8M Apr 16 21:16 audit.log
student@analystworkstation audit_logs$ ls
audit.log audit.log.1 audit.log.2 audit.log.3 audit.log.4
student@analystworkstation audit_logs$ grep -h "ssh" audit.log* | grep "res=success" | grep -o 'acct="[^\"]*"' | sort | uniq -c | sort -rn
56 acct="cbarton"
34 acct="bmorse"
student@analystworkstation audit_logs$
```

Figure 1: SSH authentication results showing cbarton (56 logins) and bmorse (34 logins).

The highlighted output in Figure 1 shows two accounts with successful SSH sessions recorded in the audit logs: cbarton with 56 successful authentication events and bmorse with 34. The grep filter on res=success ensures only successful logins are counted, excluding failed attempts. The acct= field in PAM audit records contains the authenticated username. Both accounts represent valid entry points into the system and should be scrutinized for unauthorized access.

Finding 2: User Associated with fn.invalid.html

Information Given	The filename fn.invalid.html has been flagged as suspicious by the investigation team.
Task	Identify which user is associated with opening this file.
Result	uid=1000, which corresponds to the user 'student'.

The following command was run to search for audit log entries referencing fn.invalid.html and extract the associated user ID:

```
grep -h "fn.invalid.html" audit.log* | grep -o 'uid=[0-9]*' | sort | uniq
```

```
student@analystworkstation:~/Downloads/audit_logs
student@analystworkstation audit_logs$ grep -h "fn.invalid.html" audit.log* | grep -o 'uid=[0-9]*' | sort | uniq -c
1 uid=1000
student@analystworkstation audit_logs$
```

Figure 2: Audit log entries for fn.invalid.html showing uid=1000 as the associated user.

Figure 2 shows that audit records referencing fn.invalid.html contain uid=1000. Cross-referencing with journalctl output from Figure 6, uid=1000 is confirmed to belong to the user 'student' (pam_unix logs consistently show user student(uid=1000)). This establishes that the student account was responsible for the file access event. The .invalid.html extension is not standard browser output and may indicate a redirected or injected response, making this a significant finding.

Finding 3: Evidence of .backup Directory and Scanning Tool

Information Given	Partner organizations report attackers are using a .backup directory to stage malware.
Task	Find evidence of the .backup directory in audit logs and identify the scanning tool used.
Result	/home/bmorse/.backup/ contains nmap, scan.xml, scan.nmap, spread.sh.

The following commands were used to search for PATH entries referencing .backup and extract the file names stored within:

```
grep -h "type=PATH" audit.log* | grep ".backup" | cut -d' ' -f1-10
grep -h "type=PATH" audit.log* | grep ".backup" | grep -oP 'name="\K[^\"]+' | sort |
uniq -c | sort -rn
```

```
student@analystworkstation:~/Downloads/audit_logs$ grep -h "type=PATH" audit.log* | grep ".backup" | cut -d' ' -f1-10
type=PATH msg=audit(1679075543.972:144508): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679075543.972:144509): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679075543.972:144510): item=0 name="/home/bmorse/.backup/spread.sh" inode=2322446 dev=103:01 mode=0100775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679075543.972:144511): item=0 name="/home/bmorse/.backup/nmap" inode=2322440 dev=103:01 mode=0100775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679075543.972:144512): item=0 name="/home/bmorse/.backup/scan.nmap" inode=2322442 dev=103:01 mode=0100664 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679075543.972:144513): item=0 name="/home/bmorse/.backup/scan.xml" inode=2322443 dev=103:01 mode=0100664 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679075543.972:144514): item=0 name="/home/bmorse/.backup/scan.nmap" inode=2322441 dev=103:01 mode=0100664 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001364.064:124333): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001364.064:124334): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001375.020:124337): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001375.020:124338): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001375.020:124339): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001375.020:124339): item=1 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001375.024:124342): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001375.024:124343): item=0 name="/home/bmorse/.backup/" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001375.024:124343): item=1 name="/home/bmorse/.backup/.spread.sh.swp" inode=2322445 dev=103:01 mode=0100644 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001564.691:124352): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001564.691:124353): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001579.859:124356): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001579.859:124357): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001579.859:124358): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001579.859:124358): item=1 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001579.863:124361): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001579.863:124364): item=0 name="/home/bmorse/.backup/" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001579.863:124364): item=1 name="/home/bmorse/.backup/.spread.sh.swp" inode=2322444 dev=103:01 mode=0100644 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001776.218:124391): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001776.218:124392): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001806.902:124395): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001806.902:124396): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001806.902:124397): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001806.902:124397): item=1 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001806.906:124400): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001806.906:124403): item=0 name="/home/bmorse/.backup/" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001806.906:124403): item=1 name="/home/bmorse/.backup/.spread.sh.swp" inode=2322444 dev=103:01 mode=0100644 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001875.755:124428): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001875.755:124429): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001880.099:124433): item=0 name="/home/bmorse/.backup/" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001880.099:124433): item=1 name="/home/bmorse/.backup/.spread.sh.swp" inode=2322444 dev=103:01 mode=0100644 ouid=1000 ogid=1000 rdev=00:00
type=PATH msg=audit(1679001910.228:124436): item=0 name="/home/bmorse/.backup" inode=2322439 dev=103:01 mode=040775 ouid=1000 ogid=1000 rdev=00:00
```

Figure 3: Audit PATH entries showing /home/bmorse/.backup/ with nmap, scan.xml, scan.nmap, and spread.sh.

```
student@analystworkstation audit_logs$ grep -h "type=PATH" audit.log* | grep ".backup" | grep -oP 'name="\K[^\"]+' | sort | uniq -c | sort -rn
student@analystworkstation audit_logs$
```

Figure 4: Unique path summary from the .backup directory across all audit log files.


```

student@analystworkstation:~/Downloads/audit_logs
student@analystworkstation audit_logs$ grep -h "type=EXECVE" audit.log* | grep -A2 -B2 ".backup"
student@analystworkstation audit_logs$ grep -h "type=PATH" audit.log* | grep ".backup" | grep -oP 'name="\K[^"]+' | grep -v ".backup" | sort | uniq
student@analystworkstation audit_logs$

```

Figure 5: Additional EXECVE and PATH search commands used to trace .backup activity.

Figure 3 provides the most detailed view of the .backup directory contents. Multiple audit records reference files including nmap, scan.xml, scan.nmap, spread.sh, and spread.sh.swp within /home/bmorse/.backup/. The inode and device fields confirm these are distinct physical files on the target filesystem. Nmap is the familiar scanning tool referenced in the assignment brief. The presence of scan.xml and scan.nmap suggests prior scan output was stored in the staging directory, and spread.sh indicates a shell script likely used to propagate or automate the attack. The directory owner (ouid=1000, uid=1000) initially appears to link back to uid=1000; however, the path itself is under /home/bmorse/, placing ownership with the bmorse account. This directory constitutes strong evidence of malware staging activity on the system.

Finding 4: Investigation of User 'student' via journalctl

Information Given	HR team suspects the user 'student' and requests investigation of their activity.
Task	Use journalctl to identify recent commands and activity by the student account.
Result	Student ran sudo /bin/journalctl multiple times, escalating to root.

The following commands were used to query the systemd journal for student activity:

```

sudo journalctl | grep "student" | head -50
sudo journalctl | grep "student" | grep -E "CMD|sudo|COMMAND" | head -30
sudo journalctl > /tmp/journal_export.txt
sudo grep /tmp/journal_export.txt | grep -E "CMD|sudo|session" | sort | uniq

```

```

student@analystworkstation:~/Downloads/audit_logs
student@analystworkstation audit_logs$ journalctl | grep "student" | head -50
hint: You are currently not seeing messages from other users and the system.
Users in groups 'adm', 'systemd-journal', 'wheel' can see all messages.
Pass -q to turn off this notice.
No journal files were opened due to insufficient permissions.
student@analystworkstation audit_logs$ sudo journalctl | grep "student" | head -50
[sudo] password for student:
Apr 16 21:10:51 analystworkstation systemd-logind[853]: New session 352 of user student.
Apr 16 21:10:52 analystworkstation systemd[225399]: pam_unix(systemd-user:session): session opened for user student(uid=1000) by student(uid=0)
Apr 16 21:10:52 analystworkstation systemd[1]: Started Session 352 of User student.
Apr 16 21:10:52 analystworkstation gdm-password[225357]: pam_unix(gdm-password:session): session opened for user student(uid=1000) by student(uid=0)
Apr 16 21:10:54 analystworkstation systemd[225399]: GNOME Initial Setup Copy Worker was skipped because of an unmet condition check (ConditionPathExists
! /home/student/.config/gnome-initial-setup-done).
Apr 16 21:10:56 analystworkstation systemd[225399]: GNOME Initial Setup was skipped because of an unmet condition check (ConditionPathExists=!/home/stud
ent/.config/gnome-initial-setup-done).
Apr 16 21:38:53 analystworkstation sudo[312054]: student : TTY=pts/0 ; PWD=/home/student/Downloads/audit_logs ; USER=root ; COMMAND=/bin/journalctl
Apr 16 21:38:53 analystworkstation sudo[312054]: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=1000)
student@analystworkstation audit_logs$

```

Figure 6: journalctl output showing student session events and sudo invocation.

```

student@analystworkstation audit_logs$ sudo journalctl | grep "student" | grep -E "CMD|sudo|COMMAND" | head -30
Apr 16 21:38:53 analystworkstation sudo[312054]: student : TTY=pts/0 ; PWD=/home/student/Downloads/audit_logs ; USER=root ; COMMAND=/bin/journalctl
Apr 16 21:38:53 analystworkstation sudo[312054]: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=1000)
Apr 16 21:41:41 analystworkstation sudo[317351]: student : TTY=pts/0 ; PWD=/home/student/Downloads/audit_logs ; USER=root ; COMMAND=/bin/journalctl
Apr 16 21:41:41 analystworkstation sudo[317351]: pam_unix(sudo:session): session opened for user root(uid=0) by student(uid=1000)
student@analystworkstation audit_logs$

```

Figure 7: Filtered journalctl output confirming student ran sudo /bin/journalctl with USER=root.

Indicators of Compromise Summary

The following table consolidates all indicators of compromise (IOCs) identified during the investigation:

IOC Type	Value	Context	Task
Username	cbarton	SSH login (56 sessions)	SSH Access
Username	bmorse	SSH login (34 sessions) + .backup owner	SSH Access / Malware Staging
File	fn.invalid.html	Opened by uid=1000 (student)	Suspicious File
Directory	/home/bmorse/.backup/	Malware staging directory	.backup Evidence
File	nmap (in .backup)	Network scanner stored in staging dir	Scanning Tool
File	scan.xml / scan.nmap	Nmap output files in .backup	Scanning Tool
File	spread.sh	Shell script in .backup directory	Malware Staging
Username	student (uid=1000)	Sudo activity, opened suspicious file	Student Investigation
File	status.log	Accessed 364 times (journalctl)	File Access Count

Conclusion

This investigation successfully answered all five forensic questions using standard Linux command-line tools applied to a set of audit logs and the systemd journal. Two user accounts, cbarton and bmorse, accessed the system via SSH. The suspicious file fn.invalid.html was opened by uid=1000, the student account. The /home/bmorse/.backup/ directory was confirmed as a malware staging area containing the nmap scanner, scan output files, and a shell script named spread.sh. The student account exhibited suspicious behavior by repeatedly escalating to root using sudo to run journalctl. Finally, status.log was accessed 364 times as recorded in the systemd journal.

The convergence of evidence around two accounts, bmorse (SSH access + .backup staging) and student (suspicious file access + root escalation), suggests a coordinated attack. Bmorse appears to be the primary attack account responsible for staging malicious tools, while the student account may have been compromised as a pivot point or used to perform reconnaissance within the system logs.

Recommended next steps include: (1) isolating the system immediately to prevent further compromise, (2) preserving a full forensic image of the disk before any changes are made, (3) conducting a deeper timeline analysis correlating audit log timestamps for bmorse and student activity, (4) examining spread.sh for payload delivery logic, (5) reviewing network connections associated with the nmap scans stored in .backup, and (6) conducting an account audit to determine whether cbarton's SSH access was legitimate or part of the intrusion.

References

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MITRE ATT&CK. (2023). T1078 - Valid Accounts. <https://attack.mitre.org/techniques/T1078/>

MITRE ATT&CK. (2023). T1105 - Ingress Tool Transfer. <https://attack.mitre.org/techniques/T1105/>

MITRE ATT&CK. (2023). T1083 - File and Directory Discovery. <https://attack.mitre.org/techniques/T1083/>

Appendices

Appendix A: Full Command Reference

Task 1 - SSH Users:

```
grep -h "ssh" audit.log* | grep "res=success" | grep -o 'acct="[^\"]*"' | sort |  
uniq -c | sort -rn
```

Task 2 - fn.invalid.html User:

```
grep -h "fn.invalid.html" audit.log* | grep -o 'uid=[0-9]*' | sort | uniq
```

Task 3 - .backup Directory:

```
grep -h "type=PATH" audit.log* | grep ".backup" | cut -d' ' -f1-10  
grep -h "type=PATH" audit.log* | grep ".backup" | grep -oP 'name="\K[^\"]+' | sort |  
uniq -c | sort -rn
```

Task 4 - Student Activity:

```
sudo journalctl | grep "student" | head -50  
sudo journalctl | grep "student" | grep -E "CMD|sudo|COMMAND" | head -30  
sudo journalctl > /tmp/journal_export.txt
```

Task 5 - status.log Count:

```
sudo grep "status.log" /tmp/journal_export.txt | wc -l  
grep -h "status.log" audit.log* | wc -l
```

Appendix B: MITRE ATT&CK Technique Mapping

Technique ID	Technique Name	Observed Evidence
T1078	Valid Accounts	SSH access by charton and bmorse; student sudo escalation

T1105	Ingress Tool Transfer	nmap, spread.sh, scan files staged in .backup
T1083	File and Directory Discovery	nmap scanning activity with XML/nmap output files
T1059.004	Unix Shell	spread.sh shell script found in staging directory
T1562.006	Indicator Removal: Clear Linux Logs	Student reading full journal with sudo to review evidence